Project Plan

Project Plan — OptiCrop

1. Team Members

- Mrudula Bille
- Mangala Jahnavi
- Saisri Medapureddi
- Prudhvi Pasila
- Satya Sri Geddada

Mentor: No mentor assigned

2. Project Timeline (Milestones)

Phase	Deliverable	Duration
1. Brainstorming & Ideation	Problem statement, scenarios, feasibility	Week 1
2. Requirement Analysis	SRS document	Week 1–2
3. Design Phase	Architecture, DFDs, UI design	Week 2
4. Planning Phase	Project plan, task allocation	Week 2
5. Development Phase	ML pipeline + Flask app	Week 3–5
6. Testing	Model evaluation, functional testing	Week 6
7. Documentation	Final report, user manual	Week 6–7
8. Demonstration	Demo & presentation	Week 7

3. Task Allocation (Suggested)

Team Member	Primary Responsibility
Mrudula Bille	Data preprocessing & EDA
Mangala Jahnavi	Model training & evaluation
Saisri Medapureddi	Flask backend development
Prudhvi Pasila	Frontend UI (HTML/CSS/Bootstrap)
Satya Sri Geddada	Testing, documentation & deployment

(Adjust based on actual team assignments.)

4. Resource Plan

- **Hardware**: Intel i3+, 4GB RAM minimum, 10GB free storage
- **Software**: Python 3.x, Anaconda Navigator, Jupyter Notebook/Spyder, VS Code or PyCharm, Flask
- **Dataset**: Agricultural dataset with N, P, K, temperature, humidity, pH, rainfall, and crop label

5. Risk Management

Risk	Impact	Mitigation
Dataset quality/bias	Poor prediction accuracy	Perform thorough EDA and cleaning before training
Model underperformance	Low recommendation accuracy	Compare multiple algorithms, select best performer
Timeline slippage	Delayed delivery	Weekly progress check-ins, modular development
No assigned mentor	Lack of expert guidance	Peer review within team, rely on documented best practices

6. Communication Plan

Weekly team sync to review progress against the timeline above; shared GitHub repository for version control and collaboration.